

Chapter -4 Data Movement Instruction

Machine to Assembly — 2.5 Marks

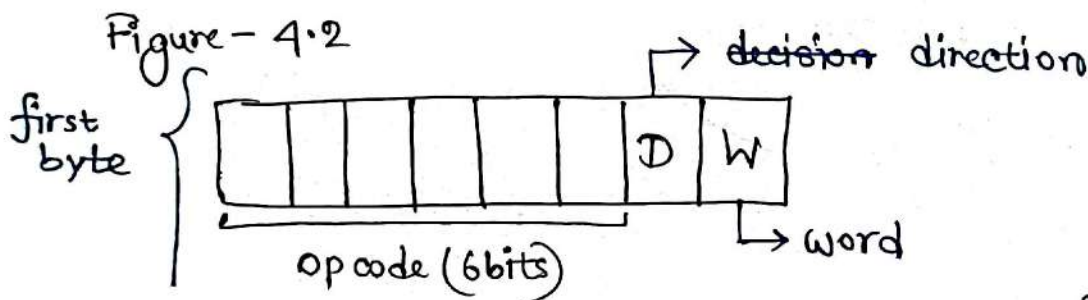
Assembly to Machine — 2.5 Marks

MOV BX, AX

8086 - Core 2

✓ (a) 16-bit form (page 132) figure 4.1-4.2 লাগবে

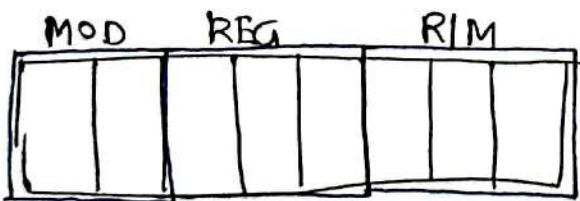
(b) 32-bit form এটা মানে রাখতে হবে



* MOV হাড়া স্থানো operation
আসে না exam ২

D = 0 $\frac{\text{dest}}{R/M} \leftarrow \frac{\text{src}}{REG}$
from REG
to R/M

Figure - 4.3



D = 1 $\frac{\text{dest}}{REG} \leftarrow \frac{\text{src}}{R/M}$
from R/M
to REG

MOD		
00	No displacement	
01	8 bit	
10	16 bit	
11	R/M is Register field	

W = $\begin{cases} 0 \rightarrow \text{byte} \\ 1 \rightarrow \text{word} \end{cases}$

Table 4.3:

REG & R/M (when)
MOD = 11

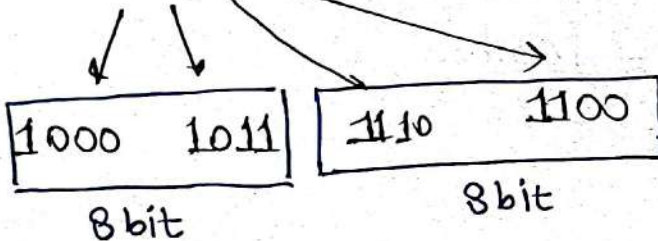
page 135

Shayan

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(figure 4.1, 4.2, 4.3) study
tables 4.1-4.2

Q. BBEC



4.2 figure

4.3 figure

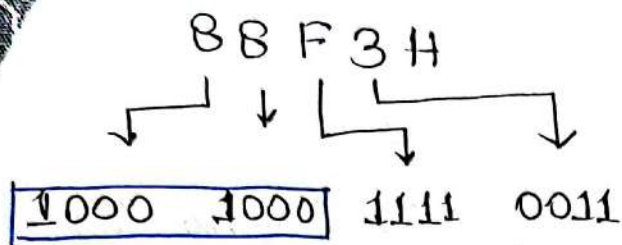
100010 → MOV r, op code

MOD = 11

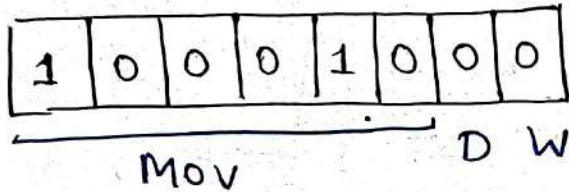
→ R/M is a register field

MOD	REG	R/M
11	101	100
	'BP'	'SP'

MOV BP, SP | Depends on direction
MOV SP, BP

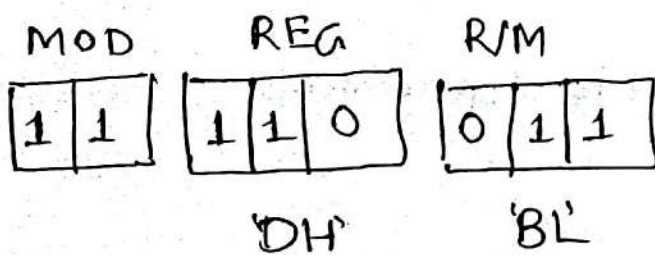


i) first byte



D = 0 R/M ← REG
W = 0 byte

ii) second byte

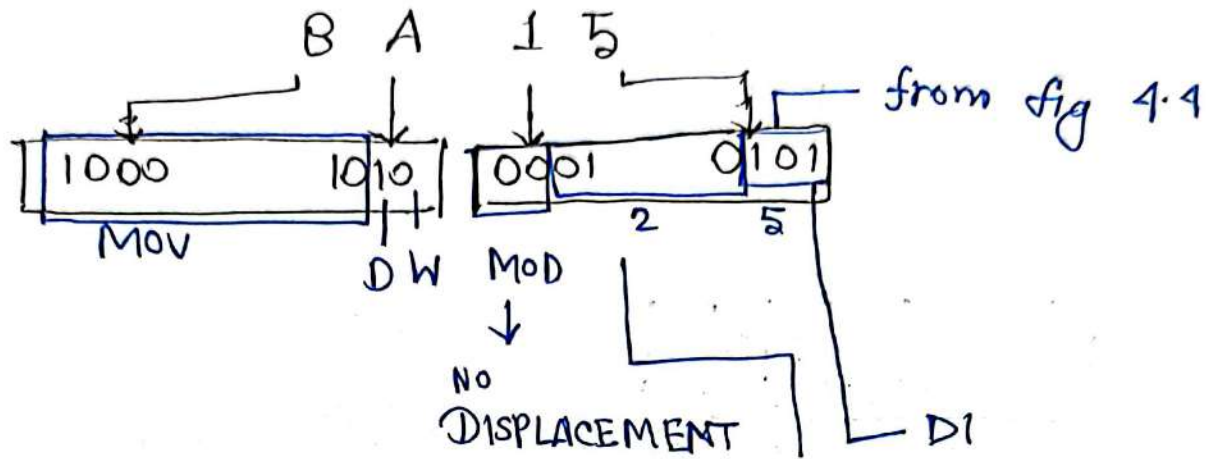


MOV BL, DH

When 00, 01, 10 is assigned to MOD.
R/M field takes on a new meaning.

* Table 4.4 | page 136

② bayan | (4)



$D \rightarrow REG \leftarrow R/M$

MOV DL, [DI]

from fig 4.3
DL

BA75 0010H

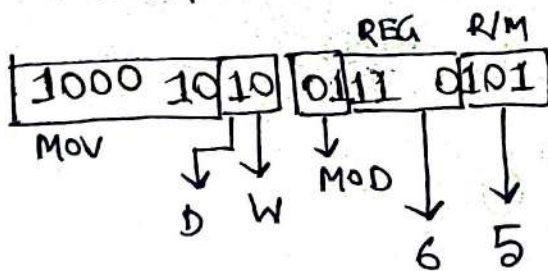
→ 16 bit displacement

BA55 01H

→ 8 bit displacement

MOV DL, [DI+01H]

BA75 0010H



$D \rightarrow REG \leftarrow R/M$

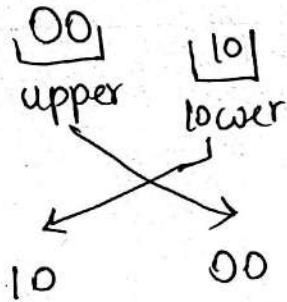
W → byte

MOD=01 → 8 bit displacement

Shayan | 5

~~80~~ 8A75 0010 H

Eta append hui reverse
16 bit hai



8AB5 0010 H

MOV DH, [DI+1000H]

Displacement low
Displacement high

BB9E 004C H

